

RW-0.003 Interface Control Document

Micro Reaction Wheel - 0.003 Nms

Hardware Revision 4

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Hardware Revision:	4
Prepared By:	AnyCompany Engineering Team
Approved By:	Systems Engineering

AnyCompany

1. Document History

Version	Date	Author	Description
1.0	2025-08-28	Engineering Team	Initial release for RW-0.003 Rev 4

1.1 Distribution List

Recipient	Organization	Copy Type
Systems Engineering	AnyCompany	Controlled
Flight Software Team	AnyCompany	Controlled
Quality Assurance	AnyCompany	Controlled
Customer Integration	External	Controlled

1.2 Reference Documents

ID	Document Number	Title
RD-1	AC-DS-00018	RW-0.003 Product Data Sheet
RD-2	AC-UM-00018	RW-0.003 User Manual
RD-3	NASA GSFC-STD-7000A	General Environmental Verification Standard (GEVS)
RD-4	ECSS-Q-ST-70-02C	Thermal Vacuum Outgassing Test
RD-5	AEC-Q200-002	Human Body Model ESD Test
RD-6	IEC 61000-4-2	Electrostatic Discharge Immunity Test
RD-7	RFC 1055	Serial Line Internet Protocol (SLIP)
RD-8	SMBus Spec v3.0	System Management Bus Specification

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3. Scope

3.1 Purpose

This Interface Control Document (ICD) defines the mechanical, electrical, environmental, and software interfaces for the AnyCompany RW-0.003 micro reaction wheel, hardware revision 4. It establishes the requirements and constraints that must be satisfied by both the reaction wheel and the host spacecraft to ensure proper integration and operation.

3.2 Product Overview

The RW-0.003 is a miniature single-axis reaction wheel providing 0.003 Nms angular momentum storage with a maximum torque of 1.5 mNm. It is designed for attitude determination and control subsystems (ADCS) on CubeSats and nanosatellites in the 1U to 6U class. The wheel operates on a 5 to 8.8 V supply (7 V nominal) and communicates via I2C or asynchronous UART interface. The unit has a mass of less than 50 g and dimensions of 33.5 x 33.5 x 17 mm.

Control modes: Speed, Torque. Qualification standard: AnyCompany internal (flight heritage qualified).

Flight heritage: 21 units with 30+ cumulative flight-years since 2023. Lifecycle status: Active, full production, estimated 8+ years to end-of-life. Typical lead time: 14 weeks.

4. Mechanical Interface

4.1 Physical Specifications

Parameter	Value	Units	Notes
Mass	< 50	g	Including connector
Dimensions (L x W x H)	33.5 x 33.5 x 17	mm	Excluding connector protrusion
Mounting	2x M2 threaded holes	-	27 mm spacing, through-holes
Min Standoff Height	3.5	mm	Boss or spacer under mount points
Mounting Hardware	M2 x 6 mm (customer)	-	Max 4 mm engagement depth
Housing Material	6061-T6 Aluminum	-	Anodized finish
Rotor Material	Stainless steel	-	Corrosion-resistant alloy

4.2 Outline Drawing and Product Photo

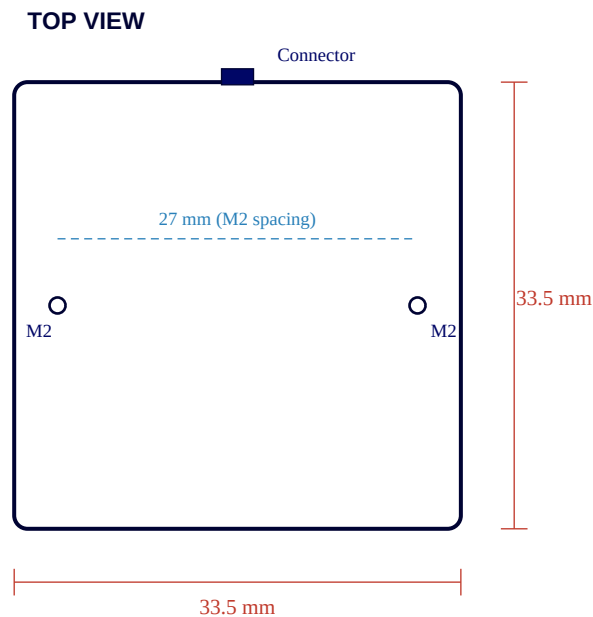


Figure 1: RW-0.003 Outline Drawing (Top View, not to scale)



Figure 2: RW-0.003 Product Photograph

4.3 Mounting Guidance

It is critical that the wheel frame not be warped when mounted as this can impede proper function. Use a spacer or boss of at least 3.5 mm beneath the two mounting points. A machined boss is the recommended approach.

4.4 Remove Before Flight Items

Item	Remove?	Notes
Kapton bearing covers	May remove	2x Kapton tape pieces, protect bearings from debris

5. Environmental Specifications

5.1 Storage

Store in a clean environment. Keep humidity low to prevent rotor corrosion. Use a sealed bag with desiccant for long-term storage.

5.2 Thermal

Parameter	Min	Typ	Max	Units	Notes
Survival Temperature	-40	-	+100	°C	Non-operating, unpowered
Operating Temperature	-30	+25	+70	°C	At baseplate interface
Qualification Range	-35	-	+75	°C	Derated from qual by 5°C
Turn-on Temperature	-30	-	+70	°C	Baseplate at power-on

Temperature is defined at the baseplate in the nominal mounting configuration.

5.3 Pressure and Outgassing

Operates in sea-level atmosphere and hard vacuum. Not qualified for high-altitude ascent; do not power during ascent without additional arcing tests. Materials: TML < 1%, CVCM < 0.1%.

5.4 Vibration

Qualified to NASA GEVS levels: 14.1 Grms for 2 minutes per axis.

Frequency [Hz]	Qual ASD [g ² /Hz]	Accept ASD [g ² /Hz]	Notes
20	0.010	0.005	Ramp-up start
50	0.040	0.020	+6 dB/oct ramp
80 - 800	0.040	0.020	Flat region
2000	0.007	0.0035	-6 dB/oct rolloff
Overall	14.1 Grms	10.0 Grms	Duration: 2 min/axis

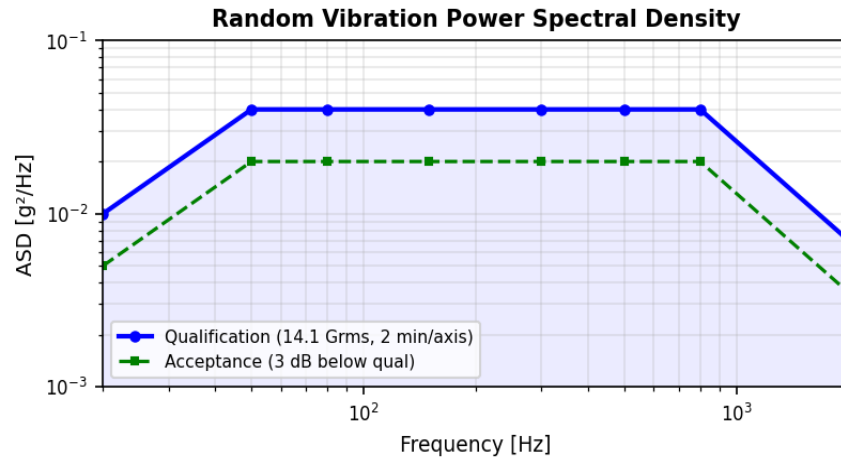


Figure 3: Random Vibration PSD Profile

5.5 Radiation

Total ionizing dose (TID) tolerance: 19.5 krad at board level. SEL immunity verified to LET > 40 MeV-cm²/mg.

6. Electrical Interface

6.1 Power Architecture

Single power input. DC/DC conversion from 5-8.8 V to internal 3.3 V. H-bridge motor driver. Regenerative braking returns energy to spacecraft bus.

V_IN: 5.0 - 8.8V (nominal 7V) | OVP threshold: 8.82V (resets wheel)

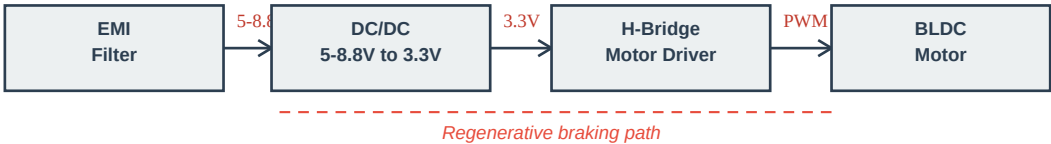


Figure 4: Power Architecture Block Diagram

6.2 Connector and Pinout

4-contact polarized nano connector. Mating: A79600-001 (Omnetics) or equivalent.

CONNECTOR FACE VIEW (Mating Side)



Figure 5: Connector Face View

Pin	Signal	Wire Color	Description	Voltage Level
1	GND	Brown	Power return, signal reference	0V
2	Power	Black	Main power input	5.0 - 8.8V (nom 7V)
3	SCL / Rx	Orange	I2C clock / UART receive	0 - 3.3V logic
4	SDA / Tx	Red	I2C data / UART transmit	0 - 3.3V logic

Table 1: Connector Pinout

6.3 Power Interface Specifications

Parameter	Min	Typ	Max	Units	Notes
Operating Voltage	5.0	7.0	8.8	V	Nominal range

Parameter	Min	Typ	Max	Units	Notes
Absolute Max Voltage	-0.3	-	11.0	V	Exceeding causes damage
OVP Reset Threshold	-	8.82	-	V	Regen triggers reset
Front-end Capacitor	-	220	-	nF	External bus capacitance
Total Bus Capacitance	-	43	-	uF	Internal + external
Inrush Current	-	-	0.5	A	At power-on, <1 ms
ESD Protection	-	-	-	-	Per HBM AEC-Q200-002

6.4 Ground Interface

Pin 1 is the return path for power and signal reference. Chassis bleed: 1 Mohm in parallel with 100 nF RF bypass cap to chassis ground plane.

6.5 Signal Interface (SDA/SCL)

Parameter	Min	Typ	Max	Units
Absolute Max Voltage	-0.5	-	5.6	V
Input High (Vih)	2.3	-	-	V
Input Low (Vil)	-	-	1.0	V
Output Low (Vol) @ 3mA	-	-	0.9	V
Pullup Current	-	0.33	-	mA
ESD Protection	-	-	-	IEC 61000-4-2, +/-30 kV

6.6 Regenerative Braking

When slowing the rotor, the wheel pushes current back onto the spacecraft bus. The spacecraft power system must accommodate this. If the power line is disconnected during regeneration, voltage rises until the 8.82 V threshold triggers a wheel reset.

7. Software Interface

7.1 Sign Convention

Positive rotation: counter-clockwise (CCW) from the top (connector side). Positive torque accelerates rotor in the positive direction.

7.2 Protocol Layer 2 - Physical

I2C: Slave with clock-stretching. Factory-programmed address. SCL held low > 25 ms triggers reset per SMBus timeout.

UART: 115200 baud, 8N1. 3.3V LVTTTL levels.

7.3 Protocol Layer 3 - Data Link (SLIP)

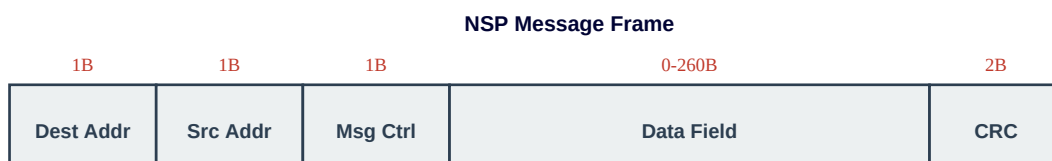
Messages framed with SLIP encoding per RFC 1055:

Character	Hex	Description
FEND	0xC0	Frame End delimiter
FESC	0xDB	Frame Escape character
TFEND	0xDC	Transposed Frame End (after FESC)
TFESC	0xDD	Transposed Frame Escape (after FESC)

FEND in data becomes FESC+TFEND. FESC in data becomes FESC+TFESC. Each message bracketed by FEND.

7.4 Protocol Layer 4 - Transport (NSP)

Nanosatellite Protocol (NSP) message format:



Max message: 265 bytes | Byte order: Little-endian | CRC-CCITT (0xFFFF init)

Figure 6: NSP Message Frame Structure

Msg control: Bit7=Poll/Final, Bit6=B flag, Bit5=ACK, Bits4-0=Command code.

CRC: polynomial $x^{16}+x^{12}+x^5+1$, init 0xFFFF, LSB first.

7.5 Protocol Layer 5 - Session (Commands)

Code	Name	Dir	Description	Data Len
0x00	PING	CMD/RPL	Heartbeat; returns status byte	0 / 1
0x01	INIT	CMD	Initialize; bootloader addr 0x00001000; no data or delay in boot	0

Code	Name	Dir	Description	Data Len
0x02	PEEK	CMD/RPL	Read memory at specified address	4 / N
0x03	POKE	CMD	Write memory at specified address	4 + N
0x04	DIAGNOSTIC	CMD/RPL	Read diagnostic channel (ADC)	1 / 2
0x06	CRC	CMD/RPL	Calculate CRC of memory region	8 / 2
0x07	READ_FILE	CMD/RPL	Read from file system (paged)	4 / N
0x08	WRITE_FILE	CMD	Write to file system (paged)	4 + N
0x09	READ_EDAC	CMD/RPL	Read EDAC-protected memory	4 / N

Table 2: Command Codes

7.6 Protocol Layer 6 - Application

Operating Modes: Bootloader and Application. Power-up starts in Bootloader. INIT with address 0x00001000 transitions to Application. INIT with no data or overvoltage returns to Bootloader.

8. Register and Memory Map

8.1 Supervisor Memory Map

Address	Field	Type	Access	Description
0x0000	speed_setpoint	int16	R/W	Commanded speed [RPM]
0x0002	torque_setpoint	int16	R/W	Commanded torque [mNm x10]
0x0004	speed_measured	int16	R	Measured speed [RPM]
0x0006	motor_current	uint16	R	Phase current [mA]
0x0008	bus_voltage	uint16	R	Input voltage [mV]
0x000A	temperature	int8	R	PCB temp [deg C]
0x000B	fault_state	uint16	R	Fault bitfield (see 9.1)
0x000D	hall_state	uint8	R	Hall sensor state
0x000E	operating_mode	uint8	R/W	0=Idle, 1=Speed, 2=Torque
0x000F	watchdog_count	uint8	R	Watchdog reset counter
0x0010	cumulative_revs	uint32	R	Total revs since manufacture
0x0014	operating_hours	uint32	R	Powered-on hours
0x0018	firmware_version	uint16	R	Firmware rev (major.minor)
0x001A	serial_number	uint32	R	Factory serial number

Table 3: Supervisor Memory Map

8.2 Diagnostic Channels

DIAGNOSTIC (0x04) with channel number in data field:

Ch	Name	Units	Scale	Range
0	wheel_speed	RPM	1 RPM/LSB	-8000 to +8000
1	motor_current	mA	0.1 mA/LSB	0 to 500
2	bus_voltage	mV	1 mV/LSB	0 to 12000
3	pcb_temperature	°C	0.1 °C/LSB	-40 to +100
4	reference_voltage	mV	1 mV/LSB	3200 to 3400

Table 4: Diagnostic Channels

8.3 EDAC Memory

256 bytes of EDAC-protected NVM (0x0000-0x00FF). Read via READ_EDAC (0x09). Single-bit errors auto-corrected; double-bit flagged in fault_state.

Address Range	Contents	Description
0x0000 - 0x001F	Calibration	Motor constants, Hall offsets, ADC gains
0x0020 - 0x003F	Configuration	I2C address, watchdog timeout, speed limits
0x0040 - 0x005F	Flight Statistics	Rev count, on-time, fault history
0x0060 - 0x007F	User Area	Customer-writable storage
0x0080 - 0x00FF	Reserved	Factory use only

9. Fault Handling

9.1 Fault State Bitfield

The `fault_state` register (0x000B, uint16). Bits set on detection, cleared by writing 0 or power cycle.

Bit	Name	Description	Recovery
0	OVERCURRENT	Motor current exceeded limit	Idle; clear fault to resume
1	OVERVOLTAGE	Bus > 8.82V	Auto-reset
2	UNDERVOLTAGE	Bus < 4.5V	Check spacecraft power
3	OVERTEMPERATURE	PCB > +80°C	Idle; cool down
4	HALL_SENSOR_FAULT	Hall readings invalid	Idle; may need service
5	COMM_TIMEOUT	No cmd within watchdog	Idle; resend commands
6	MOTOR_STALL	Cannot reach speed	Check for obstruction
7	WATCHDOG_RESET	Internal watchdog expired	Full reset; re-init
8-15	RESERVED	Reserved for future use	-

Table 5: Fault State Bitfield

9.2 Watchdog Timer

Communication watchdog: If no valid command (including PING) within 500 ms, `COMM_TIMEOUT` set, wheel enters idle. Reset by any valid command.

Hardware watchdog: Monitors firmware loop. Failure to service causes `WATCHDOG_RESET` and full wheel reset.

10. Command Modes

Application firmware supports three modes via register 0x000E or setpoint commands:

Mode	Value	Description	Entry Condition
Idle	0	Motor off, wheel coasts	Default on boot; fault recovery
Speed Control	1	PID speed loop to setpoint	Write speed_setpoint or mode=1
Torque Control	2	Open-loop torque to motor	Write torque_setpoint or mode=2

Table 6: Command Modes

Speed Control: PID-controlled speed. Torque limited to +/-1.5 mNm. Speed commands clamped to +/-8000 RPM.

Torque Control: Direct torque applied. Actual speed varies with spacecraft dynamics. Torque limited to +/-1.5 mNm.

Transitions: Immediate. Idle from spinning state coasts (no active brake). Faults force transition to Idle.

End of Document

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